

VAIBHAV GARG

Data Engineer | Data Engineer | Python | PySpark | AWS Certified | SQL | Snowflake | ETL Pipelines | Power BI

0415 735 071 | +91 74529 83652 (Indian number — reachable internationally)

vaibhavgargaaustralia@gmail.com | linkedin.com/in/vaibhavgarg003 | vaibhavgarg003.github.io

Post-Study Work Visa (Subclass 485) — 4 years remaining | No sponsorship required | Available Immediately | Open to Sydney · Melbourne · Brisbane · Perth · Adelaide

PROFESSIONAL SUMMARY

Results-driven Software and Data Engineer with 3 years of production experience designing and delivering scalable ETL/ELT pipelines, cloud-native data architectures, and ML-powered data systems using Python, PySpark, SQL, AWS, and Snowflake. AWS Certified Data Engineer – Associate (DEA-C01, 2026) — validating deep expertise across the full AWS data engineering stack including Kinesis, Glue, S3, Redshift, Athena, Step Functions, and IAM. Holds a Master of Artificial Intelligence and Machine Learning from the University of Adelaide, Australia (GPA 6.6/7.0, 2025). Delivered consistent, measurable outcomes: 30% reduction in pipeline execution time, 5+ TB of operational data processed daily with zero SLA breaches, 95%+ elimination of production data errors, and a zero-incident cloud migration. Equally strong in software engineering fundamentals — RESTful APIs, code quality, test automation, CI/CD — and in applied AI/ML across NLP, computer vision, and predictive modelling. Fully work-authorised in Australia on a Post-Study Work Visa (Subclass 485) with 4 years remaining — no sponsorship required. Available immediately.

CORE SKILLS

Languages: Python (advanced), SQL (advanced), PySpark, Java (basic), Shell Scripting

Data Engineering: ETL/ELT Pipeline Design, Apache Spark, PySpark, Hadoop, Hive, Data Ingestion, Transformation, Data Quality, Data Lineage, Medallion Architecture (Bronze/Silver/Gold), Airflow, dbt

AWS (Certified): S3, Glue, Lambda, EC2, Redshift, Kinesis, Step Functions, IAM, CloudWatch, EMR, Athena, DMS, API Gateway, SNS, SQS

Cloud & Warehousing: Snowflake, GCP, Azure (basics), Data Lakes, Dimensional Modelling, Multi-cloud Architecture

Python Ecosystem: Pandas, NumPy, Scikit-learn, SQLAlchemy, FastAPI, Flask, pytest, Matplotlib, Seaborn

ML & AI: Scikit-learn, Regression, Classification, Clustering (K-Means), Time-Series, NLP, Computer Vision (YOLOv8, OpenCV), Model Deployment

Software Engineering: RESTful API Design, Microservices, OOP, Secure Coding, Code Review, Performance Optimisation, Spring Boot

DevOps & CI/CD: Git, GitHub Actions, Docker, AWS CodePipeline, pytest — unit & integration testing, Secrets Management

BI & Monitoring: Power BI, CloudWatch, Structured Logging, SLA Tracking, Root-Cause Analysis, Runbooks

Databases: Snowflake, Redshift, MySQL, SQLite — schema design, migrations, query optimisation

PROFESSIONAL EXPERIENCE

Software Engineer | Impetus Technologies, India

Aug 2021 – Dec 2023

Python · PySpark · SQL · AWS (S3, Glue, Lambda, EC2, Redshift, Kinesis, Step Functions, CloudWatch, IAM) · Snowflake · GCP · Airflow · dbt · REST APIs · Git · Agile

- **Achieved 30% faster ETL pipeline execution processing 5+ TB daily** — re-architected distributed PySpark pipelines on AWS Glue; redesigned Medallion architecture partition strategy, eliminated shuffle-heavy joins, and parallelised Step Functions-orchestrated job chains with zero SLA breaches post-deployment.
- **Reduced production data errors by 95%+** — built Python data-quality validation frameworks enforcing schema integrity, null/duplicate detection, and referential consistency at every AWS Glue ingestion stage; zero downstream rework requests after implementation.
- **Delivered zero-incident multi-cloud migration** — migrated legacy on-premise databases to AWS (S3, Glue, Lambda) and Snowflake; designed dimensional schemas, built Python reconciliation scripts, and onboarded BI teams onto automated dashboards within 8 weeks.
- **Processed thousands of daily API requests with zero security incidents** — developed RESTful APIs (Spring Boot) with role-based authentication, input validation, IAM-equivalent policies, and structured error handling powering an ML loan eligibility scoring engine.
- **Cut pipeline failure detection from hours to under 15 minutes** — configured CloudWatch custom metrics, alarms, and log groups across production AWS environments; maintained runbooks and SOPs in Git for incident response and knowledge transfer.
- **Delivered ML solutions across Retail, Finance, and Operations verticals** — built classification, regression, time-series, and clustering models using Python and Scikit-learn; translated business problems into production-ready data science solutions for international client teams.
- **Maintained delivery quality across Agile sprints** — participated in code reviews, managed feature branches, automated test pipelines with pytest, facilitated sprint ceremonies, and prepared weekly leadership reports for international stakeholders.

Console Operator (Responsible Person) | OTR Group — Viva Energy, Adelaide

Apr 2024 – Present

Power BI · Operational Analytics · Incident Management · SLA Compliance

- **Applied Power BI in a live production environment** — monitored real-time operational KPIs, identified anomalies, applied structured resolution workflows, and maintained SLA compliance with full documentation alongside postgraduate study.
- **Managed operational incidents under pressure** — prioritised competing issues, applied SOPs, and maintained compliance standards; demonstrates structured problem-solving and accountability in a high-stakes environment.

KEY PROJECTS

Pelmorex — Multi-Cloud Data Pipeline Migration | *PySpark · AWS · Snowflake · GCP · Python · Airflow · dbt* | *Proprietary*

- **Delivered analytics data 2x faster, 30% reduction in manual operations** — migrated North American weather data platform from legacy on-premise systems to multi-cloud AWS + Snowflake + GCP using Medallion architecture; re-architected PySpark partition strategy, built Python orchestration scripts, and enforced cross-environment schema consistency with zero SLA misses.
- **Built cross-environment data quality and reconciliation layer** — Python validation enforcing integrity across AWS, Snowflake, and GCP; automated reconciliation reports with signed-off stakeholder delivery for every pipeline run.

ATM Transaction Analytics Pipeline | *PySpark · Python · SQL · Parquet · Docker · GitHub Actions · pytest* | *github.com/vaibhavgarg003*

- **12/12 pytest tests passing, full CI/CD** — production PySpark pipeline processing 10,000 transactions; time-bucket classification, high-value flagging, Parquet partitioned output, Dockerised with GitHub Actions automation.
- Football Tactical Formation Detection** | *YOLOv8 · DeepSORT · OpenCV · K-Means · Python* | *github.com/vaibhavgarg003*
- **Validated against SoccerNet-v2** — computer vision pipeline; YOLOv8 detection, HSV segmentation, DeepSORT tracking, K-Means formation classification across a full 90-minute match.

ETL Pipeline — Country Data Engineering | *Python · Pandas · SQL · SQLAlchemy · Docker · pytest* | *github.com/vaibhavgarg003*

- **15/15 tests passing, 195 countries, live REST API** — production ETL pipeline; extraction via 3rd party API, Pandas transformation, SQLite load via SQLAlchemy; Dockerised, structured logging, secrets management.

EDUCATION

Master of Artificial Intelligence & Machine Learning — University of Adelaide, Australia*Jan 2024 – Dec 2025*
GPA: 6.6/7.0 | Adelaide Graduate Award (Nov 2025) — leadership, professional excellence, cross-cultural collaboration

- Coursework: Deep learning, NLP, computer vision, reinforcement learning, time-series, ML model deployment, distributed data systems.

Bachelor of Computer Science (B.E./B.Tech) — Rajasthan Technical University 2017 – 2021
80% aggregate

CERTIFICATIONS

AWS Certified Data Engineer – Associate (DEA-C01) — Amazon Web Services, 2026 | **Advanced Microsoft Power BI** — Edureka, Mar 2026
AWS Fundamentals Specialization — Amazon Web Services, Sep 2020 | **Python for Everybody Specialization** — University of Michigan, Jul 2020

PROFESSIONAL REFEREES

Michelle Russell — Regional Manager, OTR Group | *m.russell@otr.com.au* | +61 461 283 949
Arun Nautiyal — Team Lead, Impetus Technologies | *arunnautiyal.a@gmail.com*